ONLINE STORE SALES DATA ANALYSIS

Final Project Report

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Tools: Python, Pandas, Matplotlib, Microsoft Excel, Power BI

This report presents an end-to-end analysis of online store sales data, covering data preparation, exploratory analysis, business insights, and dashboard reporting.

1. Project Overview

The project examines revenue performance, product demand, category contribution, city-wise performance, and monthly sales trends to support data-driven business decisions.

2. Objectives

• Clean and prepare the sales dataset. • Calculate revenue as Quantity × Price. • Analyze sales by product, category, city, and month. • Identify key performance indicators and business opportunities. • Present findings using charts and a dashboard.

3. Dataset and Tools

The dataset contains order-level records with Order ID, Date, Product, Category, City, Quantity, and Price. A cleaned CSV was prepared. Python/Pandas were used for cleaning and analysis; Matplotlib for charts; Excel for PivotTables/PivotCharts; and Power BI for the interactive dashboard.

4. Methodology

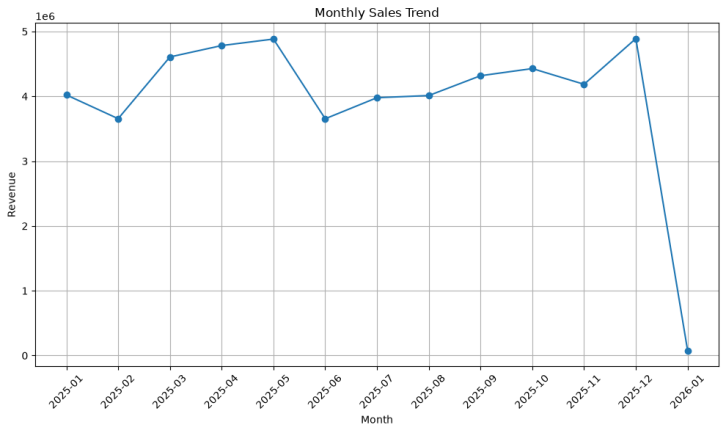
The data was loaded and inspected; missing values and duplicates were checked; text fields were standardized; dates were converted; a Revenue column was calculated; and sales were grouped by city, product, category, and month.

5. Key Performance Indicators

KPI	Result	KPI	Result
Total Revenue	■5,15,00,062	Quantity Sold	6,069
Total Orders	1,999	Average Order Value	■25,763
Total Products	12	Best-selling Product	Chair

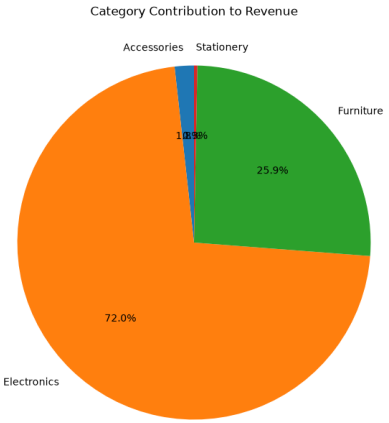
6. Sales Analysis and Visualizations

6.1 Monthly Revenue Trend



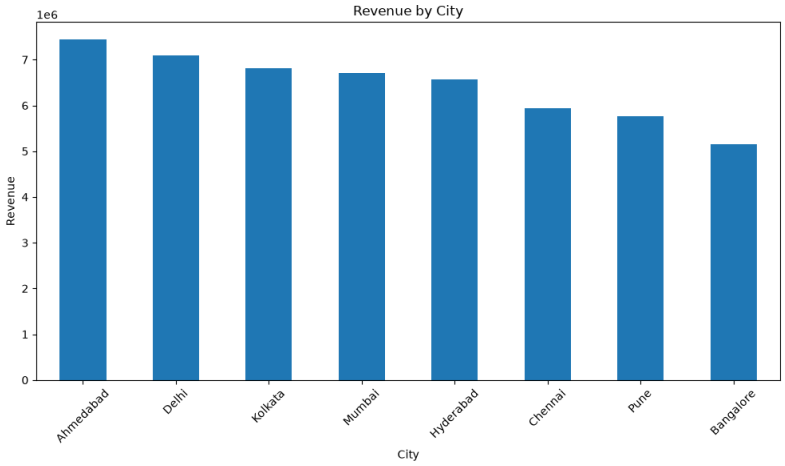
Revenue varied during 2025. May and December were among the strongest months, while February and June were comparatively lower. January 2026 contains only the records available for that month, so it should not be treated as a complete-month comparison.

6.2 Category-wise Revenue Contribution



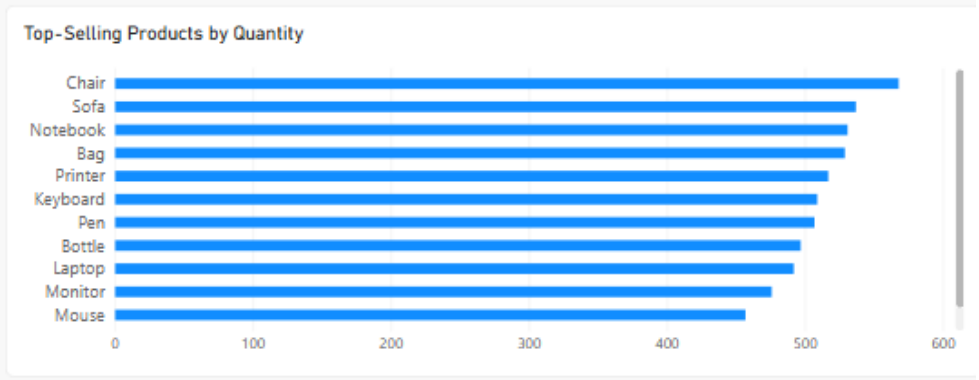
Category	Revenue	Approx. share
Electronics	■3,70,93,350	72.0%
Furniture	■1,33,43,784	25.9%
Accessories	■9,05,769	1.8%
Stationery	■1,57,159	0.3%

6.3 City-wise Revenue Analysis



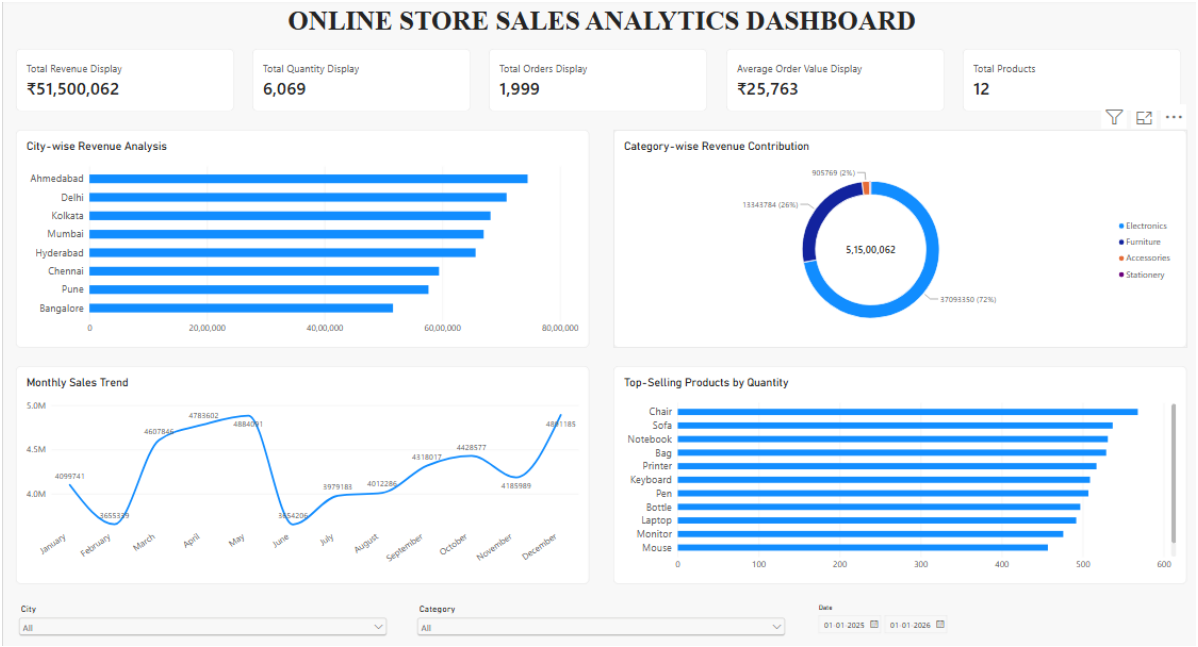
Ahmedabad has the highest revenue among the cities shown, followed by Delhi and Kolkata. Bangalore has the lowest revenue among the eight cities displayed.

6.4 Product-wise Sales Analysis



The Power BI product visual compares products by quantity sold. Chair is the best-selling product by quantity in this analysis.

7. Power BI Dashboard



The dashboard combines KPI cards with city-wise revenue, category contribution, monthly sales, and product quantity visuals. City, Category, and Date slicers allow interactive filtering.

8. Business Insights

• Electronics contributes approximately 72% of revenue, making it the leading category. • Furniture contributes approximately 26%. • Ahmedabad is the highest-revenue city in the city comparison. • Chair is the best-selling product by quantity. • Monthly revenue changes over time, so planning should account for month-to-month variation.

9. Business Recommendations

• Maintain adequate inventory for high-demand products such as chairs. • Continue supporting Electronics while exploring growth in smaller categories. • Review marketing and sales performance in lower-revenue cities. • Use monthly trends to plan promotions, inventory, and staffing. • Monitor the dashboard and investigate performance by city, category, and date.

10. Conclusion

This project demonstrates a sales analytics workflow from data cleaning and analysis to visualization and dashboard reporting. The findings highlight category performance, city-level differences, product demand, and monthly sales patterns. Python, Excel, and Power BI together provide a clear view of business performance.

GitHub Repository: Online Store Sales Analysis